

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06/E	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	K. Abhinav	Reg. No.:	192525314
Section / Batch:	2025-2029	Date:	22-7-26

Instructions to Students

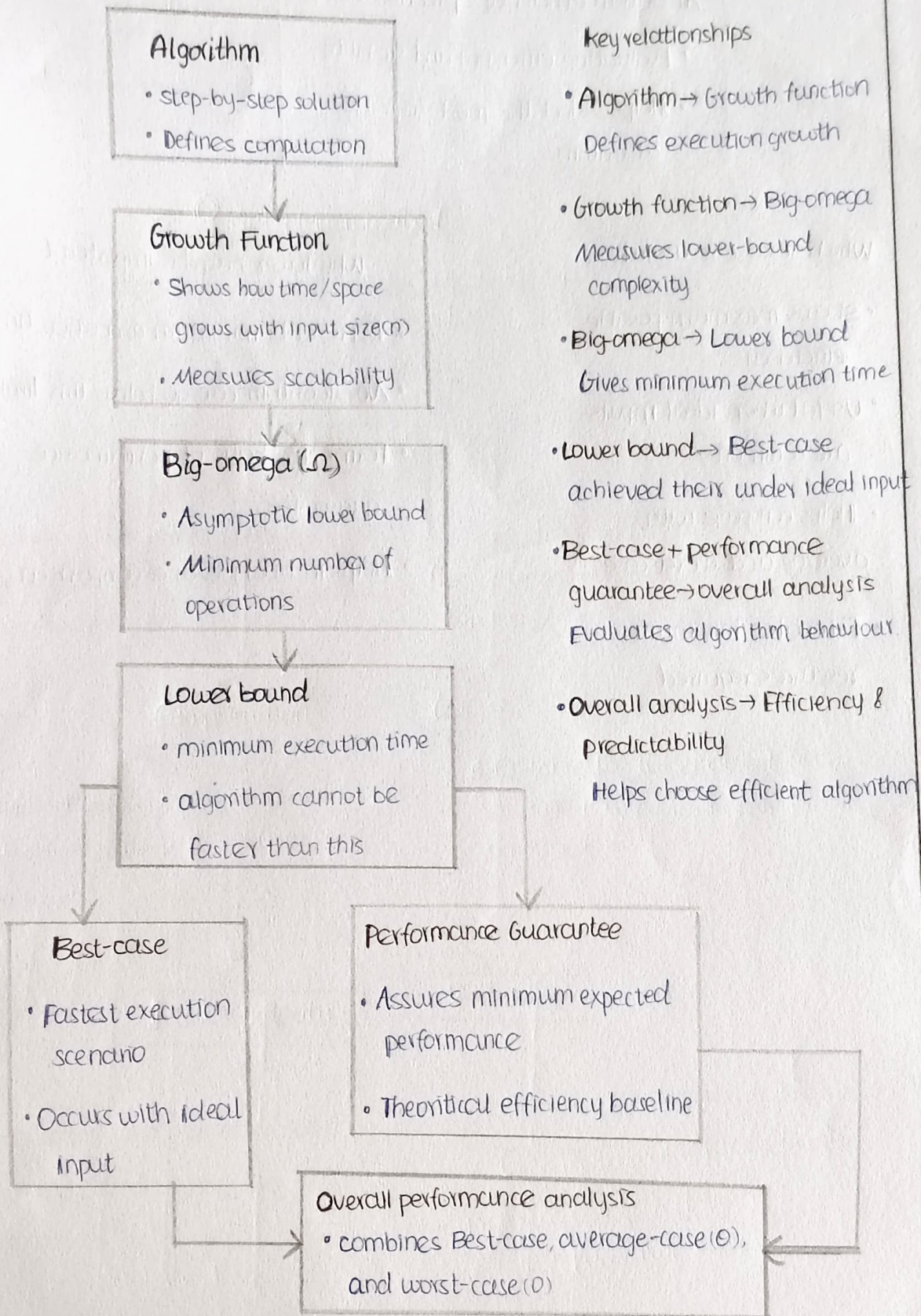
- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Consider the concept map: Algorithm → Growth Function → Big-Omega → Lower Bound → Best-case Explain how lower bound represents minimum execution time. Extend the map with performance guarantee node. Justify why best-case analysis is useful. Interpret how this contributes to overall performance understanding.

Q2. Consider the concept map: Recurrence Relation → Substitution Method → Expansion → Simplification → Final Complexity Trace the step-by-step transformation of recurrence into a closed form. Extend the map with Guess and Verification nodes. Justify how substitution helps solve non-standard recurrences. Interpret how this structured approach improves understanding.



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Algorithm efficiency & predictability

- Evaluating efficiency and stability
- Helps select the most suitable algorithm

Why best-case is useful

- Shows maximum possible efficiency.
- Useful when ideal inputs are common.
- Helps compare with average and worst case.
- Estimates minimum resources required.

Why lower bound is important

- Represents minimum execution time.
- No algorithm can go below this limit.
- Forms the basis of performance guarantee.
- Helps in analysis and comparison.

$f = \frac{n}{n}$ Recurrence Relation

Defines a mathematical equation expressing a function in terms of its values on smaller inputs

Substitution method

employs iterative algebraic unrolling to track how costs accumulate across execution levels.

Σ Expansion

generates a parameterized algebraic summation modeling the general state

Simplification

Applies boundary base cases (e.g., $n=1$) to eliminate the recursive term entirely.

Final complexity

extracts the ultimate asymptotic runtime boundary (e.g., Big-Theta notation)

leverages structural pattern recognition to formulate an educated mathematical hypothesis

Guess

establishes the target closed-form structure to be rigorously tested and confirmed

Verification

Uses mathematical induction to prove the calculated closed-form solution holds true.

Code of Conduct

I K. Abhinav (192525314) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: At

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Dr G Nagarajan		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____